17 February 2026

📊 Global E-Commerce Sales Performance & Insights Report

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Tools Used: Google Sheets, SQL, Tableau
Data Period: December 2010 – December 2011

1. Executive Summary

This study covers global sales data from December 2010 to December 2011. The raw dataset was cleaned to remove erroneous records, cancellations, and operational expenses (e.g., postage, fees) to accurately reflect true commercial performance.

2. Data Cleaning

Data Cleaning & Filtering Steps for Analysis Accuracy

- **Cancellation Filter:** Excluded all invoices starting with “C” (e.g., C536379, C536383) from the analysis.
- **Value Validation:** Ensured that Quantity and UnitPrice values are greater than zero.
- **Operational Exclusions:** Removed non-product items such as POSTAGE, Manual, DOTCOM POSTAGE, and AMAZON FEE.

3. Technical Analysis (SQL& LOGIC)

Key metric calculations from the Dataset

- **Total Revenue:** Calculated by multiplying quantity sold by unit price.
- **AOV (Average Order Value):** Determined the average spend per customer.
- **Market Ranking:** Identified top-performing products and countries by grouping revenue data.

4. Key Business Insights

A) Sales Trends

Seasonality: Sales fluctuate throughout the year, peaking notably in November–December, aligning with Black Friday and year-end campaigns.

Performance: Revenue in the second half of the year exceeds the annual average.

Insights: E-commerce revenues exhibit strong seasonality, with campaign periods accounting for a critical portion of total turnover.

B) Product Performance

Concentration: A large portion of revenue comes from a limited number of products (Top 10), with the top 3 products accounting for a disproportionately high share of total product revenue.

Category Drivers: Sales are primarily driven by decorative items and gift sets.

Insights: The sales structure is focused on “flagship” products. Cross-selling and pricing optimization present significant opportunities for these items.

C) Customer & Regional Analysis

Geographical Distribution: Revenue is primarily sourced from the United Kingdom and Western Europe.

Spending Behavior: During months with high order volume, the average basket value remains strong.

Customer Segments:

- **High-Value Customers:** Active during campaign periods with high basket values.
- **Occasional Buyers:** Engage in seasonal and promotion-driven shopping.

5. Key Final Conclusion

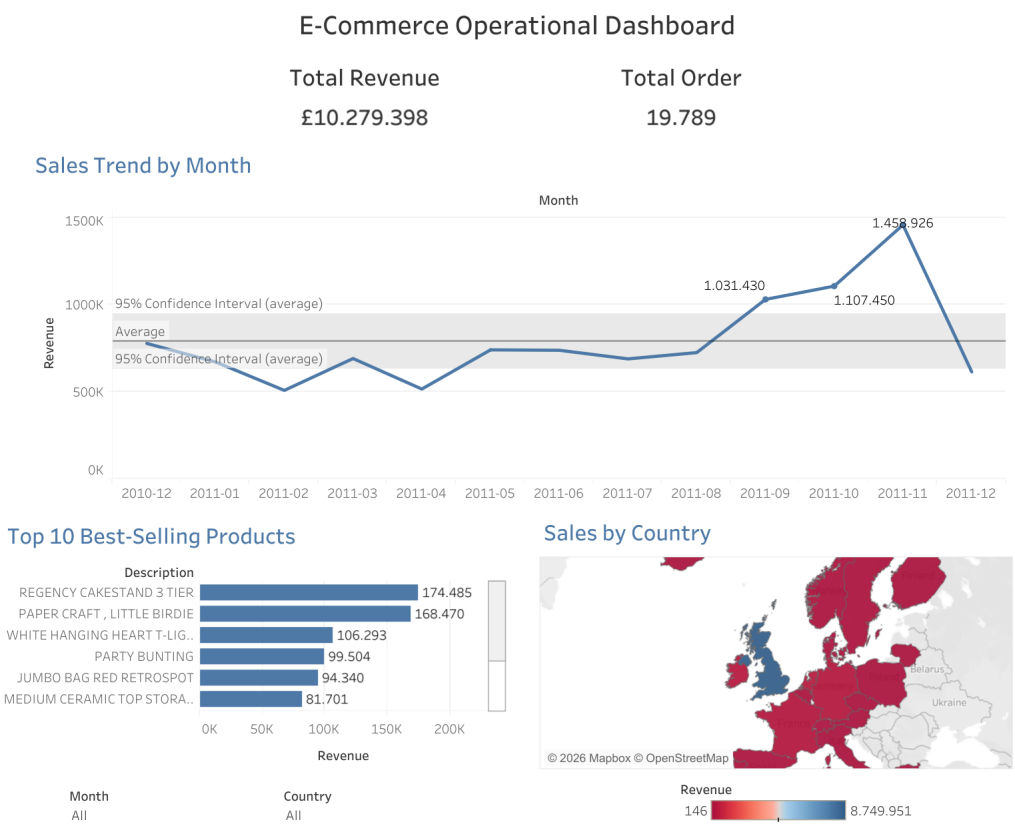
1. **Campaign Focus:** Sales performance is largely driven by seasonal campaigns.
2. **Product Concentration:** A significant portion of total revenue is concentrated in a limited number of high-performing products.
3. **Market Structure:** The main customer base is primarily located in the United Kingdom and European (UK & EU) markets.

6. Strategic Recommendations

1. **UK Market Dominance:** A large portion of the company's revenue comes from the United Kingdom; loyalty programs in this market should be strengthened.
2. **Seasonal Preparation**: Stock management should be planned starting in October to accommodate the significant demand surge during the year-end (Q4) period.
3. **Focus on Gift Sets:** Set products such as JAM MAKING SET and ALARM CLOCK have higher profitability; targeted campaigns can be executed for these items.

7. Dashboard Visualization

"Figure: Interactive Sales Dashboard Overview"



Appendix: “PostgreSQL Queries”

1)Total spending per customer

```
select "CustomerID",
ROUND(SUM("Quantity" * "UnitPrice")::numeric, 2) AS
total_spending
from kaggle
where  "Quantity" > 0
and  "UnitPrice" > 0
and  "InvoiceNo"::text not like 'C%'
and  "CustomerID" is not null
and  "Description" not in ('POSTAGE', 'Manual','DOTCOM
POSTAGE','AMAZON FEE')
group by "CustomerID"
order by total_spending desc;
```

2)Top 10 products by revenue

```
select
"Description",
round(sum("Quantity" * "UnitPrice")::numeric, 2) as revenue
from kaggle
where  "Quantity" > 0
and  "UnitPrice" > 0
and  "InvoiceNo"::text not like 'C%'
and  "Description" is not null
and  "Description" not in ('POSTAGE', 'Manual','DOTCOM POSTAGE',
'AMAZON FEE')
group by  "Description"
order by revenue DESC
LIMIT 10;
```

3)Group sales by country & date

```
select
"Country",
date_trunc('month', "InvoiceDate"::timestamp)::date AS
sale_month,
round(sum("Quantity" * "UnitPrice")::numeric, 2) AS total_sales
from kaggle
where "Quantity" > 0
and "UnitPrice" > 0
and "Description" not in ('POSTAGE', 'Manual','DOTCOM POSTAGE',
'AMAZON FEE')
and "InvoiceNo"::text not like 'C%'
and "InvoiceDate" is not null
and "Country" is not null
and "Country" not in ('Unspecified', 'Unknown')
group by "Country" , sale_month
order by "Country" , total_sales;
```